

FIU Modesto A. Maidique Campus (MMC) Virtual Tour – User Manual

Platform: Unity 2022.3 LTS

Hardware: Meta Quest 2 & Meta Quest 3

Interaction Modes: Oculus Touch Controllers & Hand Tracking

Version: 1.0

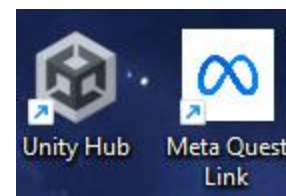
Release Date: April 2025

1. Introduction

Welcome to the immersive 360° virtual tour of FIU Modesto A. Maidique Campus. This application offers a panoramic exploration experience using inverted spheres to display high-resolution interior and exterior views. Users can interact with information pop-ups and navigate seamlessly between different campus locations.

2. System Requirements

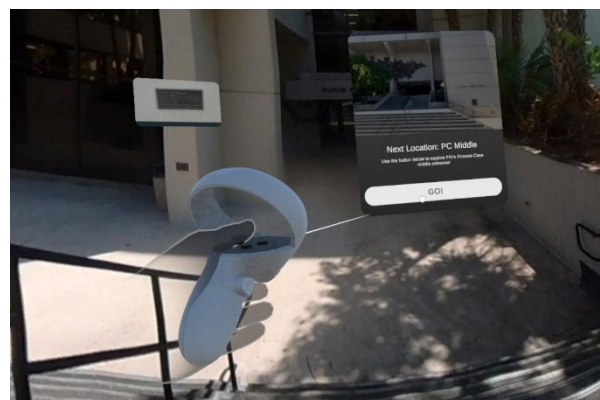
- **Headset:** Meta Quest 2 or Meta Quest 3
- **Software:** Unity VR application, Meta Quest Link



3. Controls & Interaction

3.1 Oculus Touch Controllers

- **Teleportation Between Spheres:**
 - Point at a teleportation hotspot using the controller.
 - Press the **Trigger** button to initiate teleportation.



- **Interacting with Pop-ups:**
 - Aim at an interactive element.
 - Press the **Trigger** button to engage.



- **Accessing the Scene Navigation Menu:**
 - Press the **Oculus Menu** button on the **Left Controller** to open the main navigation window.

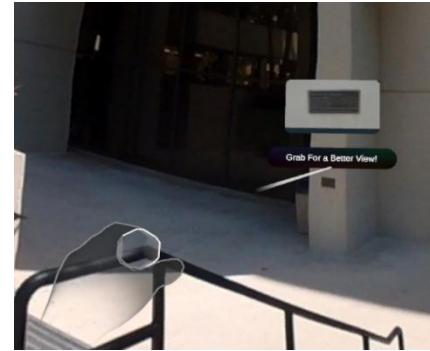


3.2 Hand Tracking

- **Teleportation Between Spheres:**
 - Reach out and perform a pinching gesture near a teleportation hotspot to initiate movement.

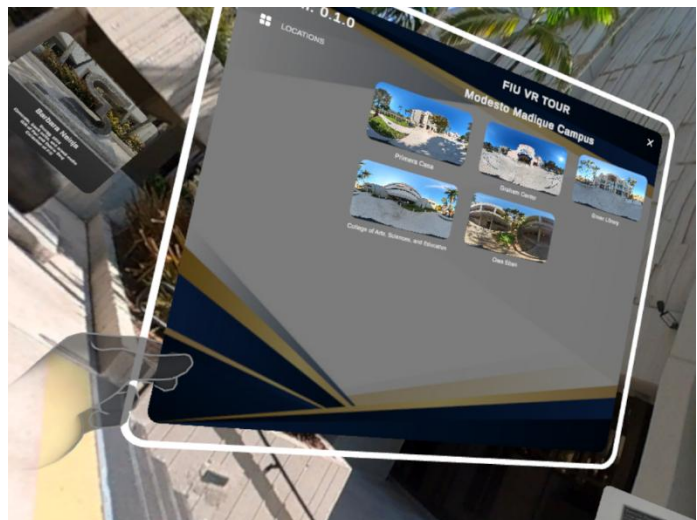
- **Interacting with Pop-ups:**

- Point at the desired pop-up and perform a grabbing gesture to interact.



- **Accessing the Scene Navigation Menu:**

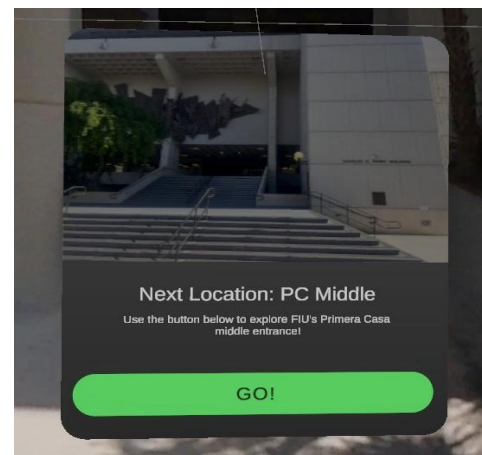
- Perform a upward pinching gesture with your **Left Hand** to open the main navigation window.



4. Navigation & Scene Management

- **Teleportation Hotspots:**

- Represented by interactive icons within each sphere.
- Engaging with these allows movement to adjacent spheres or scenes.



- **Main Navigation Window:**

- Accessible via the Oculus Menu button or left-hand pinching gesture.
- Provides a list of all available scenes for direct access.



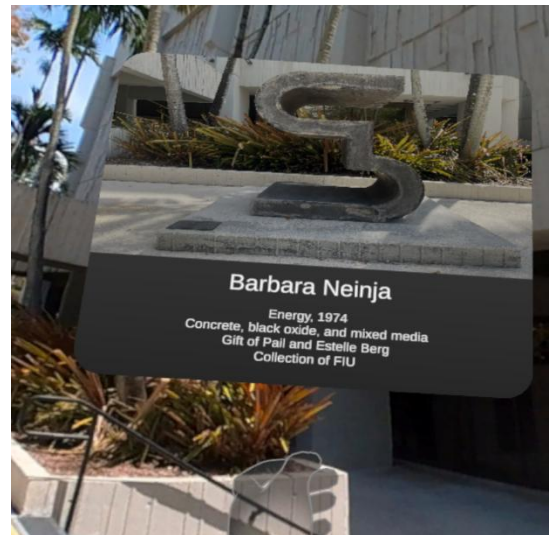
5. Information Pop-ups

- **Content:**

- Detailed descriptions of buildings, departments, and campus facilities.

- **Interaction:**

- Engage with pop-ups using the trigger button or hand gestures to read more or access related media.



6. Tips for Optimal Experience

- **Environment:**

- Use in a well-lit, open space to ensure accurate hand tracking.

- **Comfort:**

- Take regular breaks to prevent fatigue.

7. Troubleshooting

- **Hand Tracking Issues:**

- Ensure your hands are visible to the headset's cameras.

- Remove any accessories that might obstruct tracking.
 - **Navigation Menu Not Opening:**
 - Confirm you're using the correct gesture or button.
 - Restart the application if the issue persists.
 - **Pop-ups Not Responding:**
 - Check if the interactive elements are within reach.
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9. VR Hardware Troubleshooting

9.1 Black Screen on Startup

If your headset displays a black screen upon startup:

- **Charge the Headset:** Ensure the headset is fully charged using the original charger.
- **Hard Reboot:** Press and hold the power button for at least 10 seconds to perform a hard reboot.
- **Check Proximity Sensor:** Ensure the proximity sensor between the lenses is clean and unobstructed.
- **Factory Reset:** If issues persist, consider performing a factory reset via the bootloader menu. Note that this will erase all data on the device.

9.2 Elite Battery Strap Not Charging

For Meta Quest 3 users experiencing issues with the Elite Battery Strap:

- **Verify Connection:** Ensure the strap is properly connected to the headset.
- **Check for Updates:** Make sure your device firmware is up to date.
- **Contact Meta Support:** Meta has acknowledged issues with certain Elite Battery Straps and offers replacements for affected units.

9.3 Hand Tracking Not Responding

If hand tracking is unresponsive:

- **Lighting Conditions:** Use the headset in a well-lit environment to ensure optimal tracking.

- **Clean Cameras:** Gently clean the headset's cameras with a microfiber cloth.
- **Restart Headset:** Reboot the device to reset tracking systems.

9.4 Controller Connectivity Issues

For problems with controller connectivity:

- **Replace Batteries:** Ensure the controllers have fresh batteries.
- **Re-Pair Controllers:** Use the Meta Quest app to unpair and re-pair the controllers.
- **Firmware Updates:** Check for and install any available firmware updates for the controllers.

8. Support & Feedback

For assistance or to provide feedback, please contact the development team at:

virtour@fiu.edu

This user manual is designed to provide a comprehensive guide for navigating and interacting with the virtual tour application. For further customization or additional features, please consult the development documentation or contact support.